

```

=== Disaster Relief Supply Chain System ===

Dispatching supplies using strategy...
Zone 1 served: Demand=50, Urgency=5, Distance=8, Score=5
Zone 3 served: Demand=40, Urgency=7, Distance=10, Score=7
Zone 2 served: Demand=30, Urgency=8, Distance=5, Score=8

Dispatching supplies using strategy...
Zone 2 served: Demand=30, Urgency=8, Distance=5, Score=41
Zone 3 served: Demand=40, Urgency=7, Distance=10, Score=44
Zone 1 served: Demand=50, Urgency=5, Distance=8, Score=52

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Process exited after 108.4 seconds with return value 0
Press any key to continue . . . |

```

Intelligent Disaster-Relief Supply Chain

Using Graphs, Dijkstra Algorithm, Heaps and Priority Queues

5

Locations

7

Routes

8

Emergencies

0

Delivered

2

Critical

0

Response Score

Disaster Simulation

Generate emergency situations automatically and let the Priority Queue decide which emergency must be handled first.

▶ Start Disaster Simulation

▲ Generate Emergency

■ Stop Simulation

New Emergency Detected!
Destination: Village B
Required: 900 Food
Priority: 85

Interactive Disaster Transportation Graph

🚧 Block Random Road

🛠 Restore Roads

Road Blocked!
Route: Hospital C ↔ Village E
The system has removed this edge from the graph. Dijkstra will search for an alternative route.

📍 Intelligent Route Finder

Warehouse A

Hospital C

Find Shortest Route

Animate Delivery

Select locations to calculate the shortest route.

📍 Emergency Priority Queue — Max Heap

The emergency having the highest priority is always processed first.

Process Highest Priority

Reset Queue

100

Hospital C

Medicine

95

Relief Camp D

Drinking Water

100

Hospital C

Medicine

85

Village B

Food

95

Relief Camp D

Drinking Water

100

Hospital C

Medicine

100

Hospital C

Medicine

85

Village B

Food

85

Village B

Food

95

Relief Camp D

Drinking Water

95

Relief Camp D

Drinking Water

70

Village E

Blankets

95

Relief Camp D

Drinking Water

85

Village B

Food

100

Hospital C

Medicine

70

Village E

Blankets

70

Village E

Blankets

70

Village E

Blankets

85

Village B

Food

70

Village E

Blankets

85

Village B

Food

📦 Emergency Requests

Destination	Supply	Quantity	Priority	Status
Village B	Food	800	85	Waiting
Hospital C	Medicine	500	100	Waiting
Relief Camp D	Drinking Water	1000	95	Waiting
Village E	Blankets	600	70	Waiting
Village E	Blankets	500	70	Waiting
Village E	Blankets	500	70	Waiting
Village B	Food	900	85	Waiting
Village B	Food	900	85	Waiting
Relief Camp D	Drinking Water	1500	95	Waiting
Relief Camp D	Drinking Water	1500	95	Waiting
Relief Camp D	Drinking Water	1500	95	Waiting
Hospital C	Medicine	700	100	Waiting
Hospital C	Medicine	700	100	Waiting
Hospital C	Medicine	700	100	Waiting
Hospital C	Medicine	700	100	Waiting

Keller Camp D	Drinking water	1500	95	Waiting
Village B	Food	900	85	Waiting
Hospital C	Medicine	700	100	Waiting
Village E	Blankets	500	70	Waiting
Hospital C	Medicine	700	100	Waiting
Village E	Blankets	500	70	Waiting
Village E	Blankets	500	70	Waiting
Hospital C	Medicine	700	100	Waiting
Village E	Blankets	500	70	Waiting

Smart Inventory Matching

Food

500

Check Availability

Supply Available

Food: 5000 units

Required: 500 units

Remaining after delivery: 4500 units

Remaining after delivery: 4500 units

Supply	Available	Stock Level
Food	5000	
Medicine	2500	
Drinking Water	8000	
Blankets	3000	

+ Add Emergency Request

Village B

Food

500

80

Add Emergency